

# Basic concepts of the course

Mark Blumenau

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## Foreword

Knowing the answers to these questions and being able to answer 4 of them in 3 minutes guarantees the student a grade of 4 'satisfactory' for the exam. The answers provided are those that are definitely sufficient, but, of course, any correct wording and formula is accepted. In case of typos, please notify @markblumenau immediately.

## Questions

### 1. Types of gradient descent and loss functions

1. **What is mini-batch stochastic gradient descent?**

Answer: Updating model parameters based on mini-batches rather than the entire dataset.

2. **What is the idea behind gradient descent?**

Answer: Iteratively changing parameters to minimize the loss function.

3. **What is MSE?**

Answer:  $MSE = \frac{1}{n} \sum (y_{\text{pred}} - y_{\text{true}})^2$ .

4. **What loss function can be used for training in a classification task?**

Answer: Cross-entropy. (Any other correct answers are acceptable)

5. **What is the Adam optimizer?**

Answer: It combines AdaGrad and RMSProp methods for adaptive learning rate.

6. **What problem does momentum in gradient descent try to solve?**

Answer: The problem of getting stuck in shallow local minima.

7. **What method is used to calculate gradients in neural networks?**

Answer: Backpropagation.

8. **What is the parameter for weight regularization called in PyTorch?**

Answer: Weight decay.

### 2. Linear, Dropout + Batch Normalization

1. **What is a linear layer (Fully Connected Layer)?**

Answer: It is a layer described by the formula:  $y = xA^T + b$ .

2. **What is Dropout (briefly)?**

Answer: Randomly turning off neurons to prevent overfitting.

3. **What is Batch Normalization (briefly)?**

Answer: Scaling the batch at the layer input so that the mean is 0 and the variance is 1.

4. **What parameters are used in Batch Normalization?**

Answer: Parameters  $\gamma$  (scale) and  $\beta$  (bias).

5. **What is the overall goal of using Dropout and BatchNorm?**

Answer: To improve the generalization and stability of model training.

6. **How many parameters does a Linear layer have?**

Answer: (Number of inputs + 1) \* number of outputs.

7. **How does Dropout work in terms of formulas?**

Answer:  $y_{\text{dropout}} = (r \odot y)$ , where  $r \sim \text{Bernoulli}(p)$ .

8. **Write down the formulas for calculating Batch Normalization.**

Answer:

$$\mu_B = \frac{1}{m} \sum_i x_i, \quad \sigma_B^2 = \frac{1}{m} \sum_i (x_i - \mu_B)^2, \quad \hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}, \quad \text{quad } y_i = \gamma \hat{x}_i + \beta.$$

9. **What role does the ReLU activation function play?**

Answer: It introduces nonlinearity:  $f(x) = \max(0, x)$ .

10. **Why is nonlinearity between layers necessary?**

Answer: Using two linear layers in a row is equivalent to using a single linear layer.

11. **What is the Universal Approximation Theorem? (Tsybko)**

Answer: A linear neural network with one hidden layer can approximate any continuous function of many variables with any accuracy.

### 3. Convolution operation

1. **What is a convolution layer in CNN?**

Answer: It performs image convolution with filters to extract features.

2. **How many parameters are there in a Conv2d layer?**

Answer:  $K \cdot K \cdot C_{in} \cdot C_{out} + C_{out}$ , where  $K$  is the kernel size and  $C$  is the number of channels.

3. **How many multiplications are there in the Conv2d layer?**

Answer:  $H \cdot W \cdot C_{out} \cdot (K \cdot K \cdot C_{in})$ , where  $K$  is the kernel size,  $C$  is the number of channels,  $H$  and  $W$  are the height and width of the output tensor.

4. **What is the role of the convolution kernel?**

Answer: It highlights local patterns through weighted summation of pixels.

5. **What is stride in a convolution layer?**

Answer: The number of pixels by which the filter shifts during convolution.

6. **What is padding in a convolution layer?**

Answer: Adding extra pixels to preserve dimensionality.

7. **What does the pooling operation do?**

Answer: It reduces the dimension by taking a given function in a given window.

8. **How is the size of the output tensor of a convolution layer calculated?**

Answer:  $O = \frac{W-F+2P}{S} + 1$ , where  $W$  — input size,  $F$  — kernel size,  $P$  — padding,  $S$  — stride.

9. **Write down the formula for the convolution operation for an image.**

OTBET:

$$\text{Im}^{\text{out}}(x, y, t) = \sum_{i=-d}^d \sum_{j=-d}^d \sum_{c=1}^c (K_t(i, j, c) \text{Im}^{\text{in}}(x+i, y+j, c) + b_t)$$

10. **How does the dimension change with max pooling?**

Answer: The dimension =  $\frac{W-F}{S} + 1$ , where  $F$  is the window size and  $S$  is the pooling stride.

11. **What is zero-padding and what is it used for?**

Answer: Adding zero elements at the edges to preserve the dimension after convolution.

## 4. CNN Architectures

1. **What characterizes the VGG architecture?**

Answer: A deep network with sequential convolutional layers and small filters.

2. **Write down the formula for the residual block in ResNet.**

Answer:  $y = F(x, \{W_i\}) + x$ .

3. **What is ResNet?**

Answer: A network with residual connections that allow for the construction of deeper models.

## 5. Computer vision tasks

1. **What is image classification?**

Answer: Determining the category of an object in an image.

2. **What is object detection?**

Answer: Identifying and locating objects, specifying their coordinates and category.

3. **What is image segmentation?**

Answer: Dividing an image into regions corresponding to different objects.

4. **Write down the formula for calculating IoU in object detection.**

Answer:  $\text{IoU} = \frac{\text{area}(\text{intersection})}{\text{area}(\text{union})}$ .

5. **How is Non-Maximum Suppression (NMS) applied in object detection?**

Answer: Boxes with the highest confidence are selected, removing those that overlap significantly with them.

6. **How is the image segmentation task formalized?**

Answer: The function  $L(\text{mask}_{\text{pred}}, \text{mask}_{\text{true}})$  is minimized, for example, dice loss.

7. **How is the mAP metric defined for detection evaluation?**

Answer: By averaging AP across classes.  $\text{mAP} = \frac{1}{N} \sum \text{AP}_i$ , where AP is the area under the precision-recall curve.

## 6. Embeddings

### 1. What are embeddings?

Answer: Representation of discrete data as dense vector spaces.

### 2. What is cosine similarity for embeddings?

Answer:  $\cos(\theta) = \frac{A \cdot B}{\|A\| \cdot \|B\|}$ .

### 3. What does the softmax function for embedding normalization look like?

Answer:  $\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$ .

## 7. RNN, LSTM, GRU

### 1. What is an RNN?

Answer: A recurrent neural network that processes sequential data while preserving state (in what is known as a hidden state).

### 2. What is an LSTM?

Answer: An RNN with memory cells and gates to control the flow of information, preventing gradient vanishing.

### 3. What is a GRU?

Answer: A simplified version of an LSTM with two gates (update and reset).

### 4. Write down the formula for updating the hidden state in a standard RNN.

Answer:

$$h_t = \tanh(x_t W_{ih}^T + b_{ih} + h_{t-1} W_{hh}^T + b_{hh})$$

## 8. Generative models

### 1. What is latent space?

Answer: It is a hidden representation of data where its key features are stored compactly.

### 2. What is an autoencoder and what is its structure?

Answer: An autoencoder is a network that compresses data into a hidden space and reconstructs it back. It consists of an encoder and a decoder.

### 3. How does the decoder work in an autoencoder?

Answer: It takes the latent vector and converts it back into an image.

### 4. What is a generative adversarial network (GAN)?

Answer: It is a model where a generator creates images and a discriminator tries to distinguish them from real ones.

### 5. Explain the term mode collapse. For which family of generative models is mode collapse characteristic?

Answer: Mode collapse is when the generator starts to produce a very limited variety of results. It is characteristic of GANs.

### 6. Why is GAN often unstable during training?

Answer: Because of the competitive process and sensitivity to the balance between networks.

### 7. What is a diffusion model?

Answer: A model that has been trained to remove noise step by step, restoring the image.

8. **How is the process of adding noise to an image organized when training a diffusion model?**

Answer: Noise is added gradually, bringing the image closer to pure noise.

9. **What are normalization flows?**

Answer: Reversible transformations that model complex data distributions.

10. **What are the main limitations of normalization flows?**

Answer: They are computationally heavy and require strict reversibility.

11. **What is the disadvantage of diffusion models in terms of computational cost?**

Answer: They require many steps, which makes generation slow.

## 9. Transformers

1. **What is the attention mechanism?**

Answer: Attention is a way to highlight the most important parts of the input for the current processing step.

2. **What do the terms Query, Key, and Value mean?**

Answer: These are three representations of tokens through which attention is calculated.

- Query (request) - what am I looking for?
- Key (key) - what am I offering?
- Value - what will I get if the match is successful?

3. **How is the attention score calculated?**

Answer:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

4. **Why do multiple attention heads improve the accuracy of transformer models (compared to a single attention head)?**

Answer: Because different heads focus on different patterns in the data.

5. **What does position encoding do and why is it needed?**

Answer: It adds information about the position of tokens, which a regular transformer lacks by default.

6. **What does masked self-attention do?**

Answer: It prevents looking at future tokens during generation.

7. **How does GPT differ from BERT?**

Answer: GPT is a decoder, BERT is an encoder.

8. **What is LoRA?**

Answer: LoRA (Low-Rank Adaptation) is a method of retraining large neural networks in which the main weights of the model are frozen and training proceeds only on small low-rank additions.